Learning Noisy Labels



Noise Exists

- Problems with sensors



- Semantic ambiguity sometimes our defined semantic classes overlap

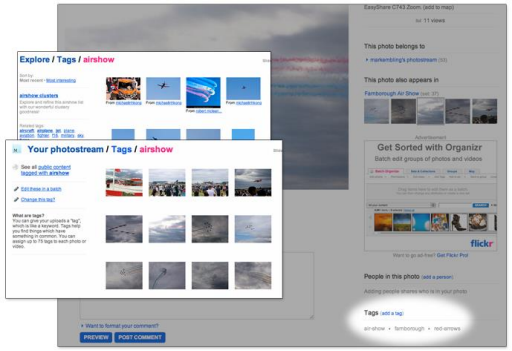


- Human error



- one way to decrease annotation effort is to automatically collect (noisy) labels

[Sun et al., "Revisiting Unreasonable Effectiveness of Data in Deep Learning Era", ICCV 2017]



Robust Loss Function

- Train using a loss function that is provably robust to label noise of different sorts
- Symmetric noise
  - Same noise pattern across classes
- Asymmetric noise (class conditional)
  - Different noise pattern depending on the true class
- Variant noise (sample conditional)
  - Different noise pattern depending on the true class and the sample

[Ghosh et al,. “Robust Loss Functions under Label Noise for Deep Neural Networks”, AAAI 2017]

Robust Loss Function

- Train using a loss function that is provably robust to label noise of different sorts
- Symmetric noise
  - Same noise pattern across classes
- Asymmetric noise (class conditional)
  - Different noise pattern depending on the true class
- Variant noise (sample conditional)
  - Different noise pattern depending on the true class and the sample

[Ghosh et al,. “Robust Loss Functions under Label Noise for Deep Neural Networks”, AAAI 2017]



Identify and Remove

- Identify samples that are potentially noisy
  - use high learning rate, take the samples that are not classified correctly
  - choose samples based on the amount of time it takes to correctly classify the sample
  - choose samples based on the variance of the predicted labels
  - choose samples based on the uncertainty of the model prediction (e.g., use ensembles)
  - …
- Remove them from the training set
- Retrain

[Huang et al., "O2U-Net: A Simple Noisy Label Detection Approach for Deep Neural Networks", ICCV 2019]



Joint Identification and Re-Labeling

- Identify samples that are potentially noisy
  - as in the previous slide
- Relabel those samples using some heuristics
  - Approaches from semi-supervised learning such as pseudo labeling
- Retrain the model
- Reiterate from start

[Tanaka et al., “Joint Optimization Framework for Learning with Noisy Labels”, CVPR 2018]

Premature Early Stopping

- Deep networks usually learn from clean data first and the adapt to noise towards the end of training
  - “A Closer Look at Memorization in Deep Networks”, ICML 2017
  - Liu et al. “Early-Learning Regularization Prevents Memorization of Noisy Labels”, NeurIPS 2020

